

2 COMPONENTS

code

Code — syntax-highlighted source code blocks.

How a signal earns its score.

```
function scoreSignal({ confidence, relevance, observedAt }, weights) {  
  const ageDays = (Date.now() - observedAt) / DAY_MS;  
  const recency = Math.exp(-ageDays / weights.halfLife);  
  
  return (  
    confidence * weights.confidence +  
    recency * weights.recency +  
    relevance * weights.relevance  
  );  
}
```

The N+1 query that slowed every report.

BEFORE · ONE QUERY PER ROW

```
const signals = await db.signals.findAll();
for (const s of signals) {
  s.owner = await db.users.find(s.ownerId);
}
return signals;
```

AFTER · ONE BATCHED JOIN

```
const signals = await db.signals.findAll({
  include: { owner: true },
});
return signals;
```